TAUTNESS OF AN ARROW OF COMPLETE WEAKLY ADMISSIBLE RINGS IS A LOCAL PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof opens by observing that the locality axioms have already been obtained from a geometric statement, and then gives a direct algebraic proof of two of them; the final sentence is the author's 'We omit the proof of (axiom-3).' That sentence is reproduced verbatim and recorded as an obligation, and it is to be read together with the geometric argument earlier in the same span, which does cover the third axiom.

Editorial note. The proof displays a commutative diagram of quotients and a three-line chain of equalities computing an intersection; both lie inside the counted span and are reproduced verbatim.

Editorial note. The statement refers to the notions of tautness and to the locality axioms by label; those references are carried as navigation stubs and the definitions are recorded as prerequisites rather than reproduced.

Editorial note. The counted spans carry 12 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Tautness of an arrow of complete weakly admissible rings, and complete weakly admissible rings themselves. Same chapter, Definition definition-taut, invoked in the statement at line 5469. Fixes the property whose equivalence with the inverse system formulation and whose locality is asserted.
- The translation of the equivalent conditions into representability of the corresponding morphism, and locality of representability. Same chapter, Lemma lemma-representable-affine at proof lines 5492-5494 and Lemma lemma-representable-by-algebraic-spaces-local at proof lines 5495-5497. Gives all three locality axioms by the geometric route, before the algebraic proof is given.
- The situation for a local property with its axioms, and the example showing the comparison arrows satisfy the condition. Same chapter, Situation situation-local-property and Equation equation-localize referred to at

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proof lines 5500-5502, and Example example-representable-morphism-from-completion at proof lines 5502-5504. Fixes the shape of the diagram the algebraic proof works with and discharges the case of the comparison arrows.

- The closure of a weak ideal of definition, and topological nilpotence of elements of such closures. Same chapter, Lemma lemma-closed at proof lines 5541-5542 and Lemma lemma-topologically-nilpotent at proof lines 5544-5546. Identify the sets produced by the computation as closures of ideals and show they are weak ideals of definition.
- Faithful flatness implies that an intersection can be tested after base change. Algebra, Lemma algebra-lemma-faithfully-flat-universally-injective, invoked at proof lines 5537-5539. Gives the third equality in the computation for the second axiom.

Source reference. Original lemma 05CK: sources/stacks-chow/tags.txt, source lines 6923–6923. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original equation 0ANC: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5063–5063. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 0AN6: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4393–4393. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original reference 0AND: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5086–5086. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original reference 0ANE: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5090–5090. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original reference 0ANF: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5095–5095. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AMS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 922–922. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN5: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4239–4239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN4: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4094–4094. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AMW: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1134–1134. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0ANB: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5029–5029. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 0CBA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5059–5059. Pointer outside the counted span, invoked only from a delivered context span as a forward reference; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let $\varphi : A \to B$ be a continuous map of linearly topologized rings. We say φ is *taut*[1] if for every open ideal $I \subset A$ the closure of the ideal $\varphi(I)B$ is open and these closures form a fundamental system of open ideals.

Lemma 0.2. *Let $B \to A$ be an arrow of $WAdm^{count}$. The following are equivalent*

(a) *$B \to A$ is taut (Definition 0.1),*
(b) *for $B \supset J_1 \supset J_2 \supset J_3 \supset \ldots$ a fundamental system of weak ideals of definitions there exist a commutative diagram*

$$\begin{array}{ccccccccc} A & \longrightarrow & \ldots & \longrightarrow & A_3 & \longrightarrow & A_2 & \longrightarrow & A_1 \\ \uparrow & & & & \uparrow & & \uparrow & & \uparrow \\ B & \longrightarrow & \ldots & \longrightarrow & B/J_3 & \longrightarrow & B/J_2 & \longrightarrow & B/J_1 \end{array}$$

such that $A_{n+1}/J_nA_{n+1} = A_n$ and $A = \lim A_n$ as topological ring.

Moreover, these equivalent conditions define a local property, i.e., they satisfy axioms (0AND), (0ANE), (0ANF).

Proof. The equivalence of (a) and (b) is immediate. Below we will give an algebraic proof of the axioms, but it turns out we've already proven them. Namely, using Lemma 0AN5 the equivalent conditions (a) and (b) translate to saying the corresponding morphism of affine formal algebraic spaces is representable by algebraic spaces. Since this condition is "étale local on the source and target" by Lemma 0AN4 we immediately get axioms (0AND), (0ANE), and (0ANF).

Direct algebraic proof of (0AND), (0ANE), (0ANF). Let a diagram (0ANC) as in Situation 0CBA be given. By Example 0AN6 the maps $A \to (A')^\wedge$ and $B \to (B')^\wedge$ satisfy (a) and (b).

[1]This is nonstandard notation. The definition generalizes to modules, by saying a linearly topologized A-module M is A-taut if for every open ideal $I \subset A$ the closure of IM in M is open and these closures form a fundamental system of neighbourhoods of 0 in M.

Assume (a) and (b) hold for φ. Let $J \subset B$ be a weak ideal of definition. Then the closure of JA, resp. $J(B')^\wedge$ is a weak ideal of definition $I \subset A$, resp. $J' \subset (B')^\wedge$. Then the closure of $I(A')^\wedge$ is a weak ideal of definition $I' \subset (A')^\wedge$. A topological argument shows that I' is also the closure of $J(A')^\wedge$ and of $J'(A')^\wedge$. Finally, as J runs over a fundamental system of weak ideals of definition of B so do the ideals I and I' in A and $(A')^\wedge$. It follows that (a) holds for φ'. This proves (0AND).

Assume $A \to A'$ is faithfully flat and that (a) and (b) hold for φ'. Let $J \subset B$ be a weak ideal of definition. Using (a) and (b) for the maps $B \to (B')^\wedge \to (A')^\wedge$ we find that the closure I' of $J(A')^\wedge$ is a weak ideal of definition. In particular, I' is open and hence the inverse image of I' in A is open. Now we have (explanation below)

$$\begin{aligned} A \cap I' &= A \cap \bigcap (J(A')^\wedge + \mathrm{Ker}((A')^\wedge \to A'/I_0A')) \\ &= A \cap \bigcap \mathrm{Ker}((A')^\wedge \to A'/JA' + I_0A') \\ &= \bigcap (JA + I_0) \end{aligned}$$

which is the closure of JA by Lemma 0AMS. The intersections are over weak ideals of definition $I_0 \subset A$. The first equality because a fundamental system of neighbourhoods of 0 in $(A')^\wedge$ are the kernels of the maps $(A')^\wedge \to A'/I_0A'$. The second equality is trivial. The third equality because $A \to A'$ is faithfully flat, see Algebra, Lemma 05CK. Thus the closure of JA is open. By Lemma 0AMW the closure of JA is a weak ideal of definition of A. Finally, given a weak ideal of definition $I \subset A$ we can find J such that $J(A')^\wedge$ is contained in the closure of $I(A')^\wedge$ by property (a) for $B \to (B')^\wedge$ and φ'. Thus we see that (a) holds for φ. This proves (0ANE).

We omit the proof of (0ANF). $\square$